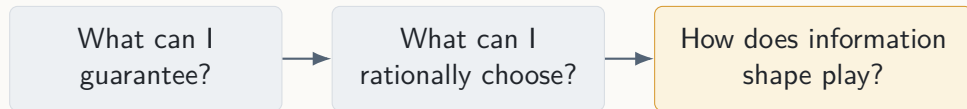


Advanced Game Theory

Information and rational behavior

Lecture 2



Olivier Gossner

From knowledge to choices

Lecture 1

Information cells
Conditional beliefs
Knowledge about others

Lecture 2

Beliefs about actions
Best responses
Rationality about others

Which choices can information and rationality support?

Four lectures

- 1 Knowledge and agreement**
Partitions; common knowledge; Aumann's theorem
- 2 Information and rational behavior**
Minimax; rationalizability; correlated equilibrium; Bayesian games
- 3 Comparing and choosing information**
Blackwell; strategic value; information design
- 4 Higher-order beliefs and universal representation**
Email game; universal type space; interim rationalizability

1

Minimax

Guarantees; mixing; a separation argument

2

Rationalizability

Best responses; conjectures; iterating rationality

3

Correlated equilibrium

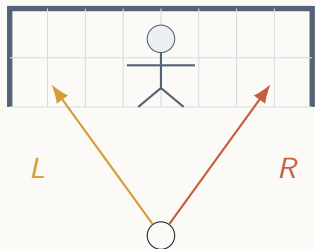
Signals; obedience; the revelation principle

4

Bayesian games

Types; contingent choices; equilibrium

Guarantees: penalty shootouts



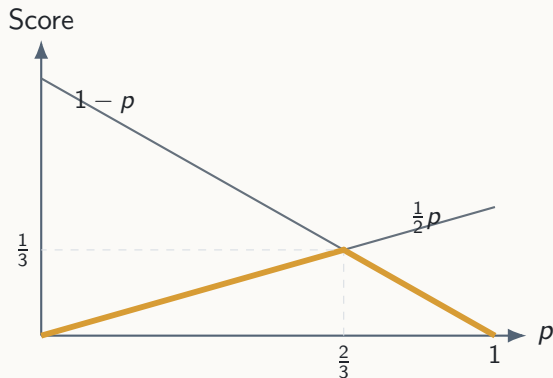
	l	r
L	0	α_L
R	α_R	0

Shooter's scoring probability

$$\alpha_L > \alpha_R > 0$$

Choose a distribution before seeing the opponent's action.

The shooter maximizes a guarantee



$$p = \Pr(R)$$

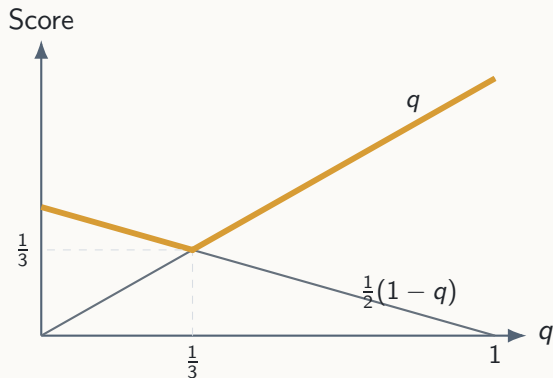
$$\min\{\alpha_R p, \alpha_L(1-p)\}.$$

Illustration:

$$\alpha_L = 1, \alpha_R = \frac{1}{2}.$$

Mix until the goalkeeper's two responses give the same score.

The goalkeeper minimizes the worst case



$$q = \Pr(r)$$

$$\max\{\alpha_L q, \alpha_R(1-q)\}.$$

Same illustration:

$$\alpha_L = 1, \alpha_R = \frac{1}{2}.$$

Mix until the shooter's two actions give the same score.

Two guarantees meet

Shooter

$$p^* = \frac{\alpha_L}{\alpha_L + \alpha_R}$$

At least v against every goalkeeper.

Goalkeeper

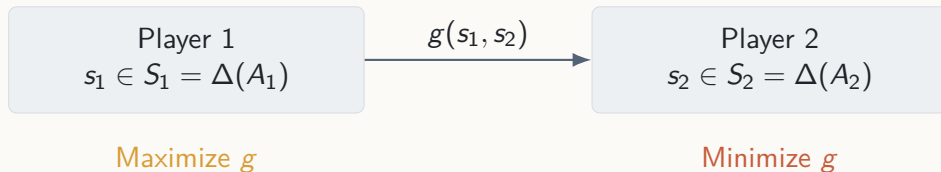
$$q^* = \frac{\alpha_R}{\alpha_L + \alpha_R}$$

At most v against every shooter.

$$v = \frac{\alpha_L \alpha_R}{\alpha_L + \alpha_R}$$

The less effective kick is used more often.

Minimax: the general zero-sum game



$$g(s_1, s_2) = \sum_{a_1, a_2} s_1(a_1) s_2(a_2) g(a_1, a_2).$$

Finite action sets; payoffs g and $-g$.

Two orders of choice

Player 1 commits first

$$\underline{v} = \max_{s_1} \min_{s_2} g(s_1, s_2)$$

The best guarantee.

Player 2 commits first

$$\overline{v} = \min_{s_2} \max_{s_1} g(s_1, s_2)$$

The smallest upper bound.

$$\underline{v} \leq \overline{v} \quad \text{Can mixing close the gap?}$$

The minimax theorem

von Neumann (1928)

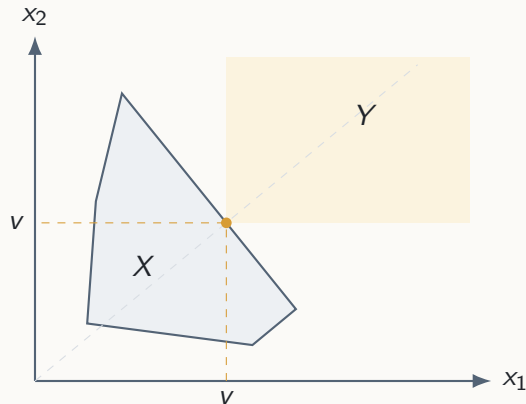
$$\max_{s_1 \in S_1} \min_{s_2 \in S_2} g(s_1, s_2) = \min_{s_2 \in S_2} \max_{s_1 \in S_1} g(s_1, s_2) = v.$$

Optimal strategies s_1^*, s_2^* satisfy

$$g(s_1, s_2^*) \leq v \leq g(s_1^*, s_2) \quad \text{for every } s_1, s_2.$$

(s_1^*, s_2^*) is a Nash equilibrium.

Proof: payoff vectors and an unattainable region



One coordinate per action of Player 2:

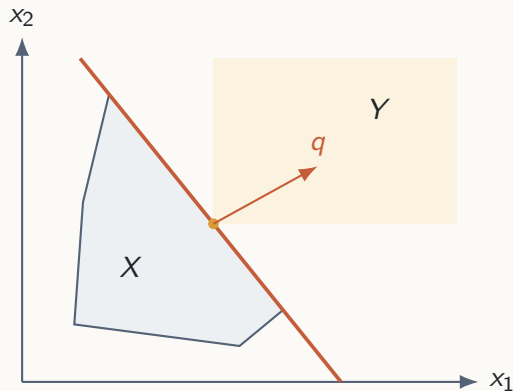
$$X = \left\{ (g(s_1, j))_{j \in A_2} : s_1 \in S_1 \right\}.$$

$$v = \max_{x \in X} \min_j x_j.$$

$$Y = \{y : y_j > v \text{ for all } j\}.$$

X is compact and convex; $X \cap Y = \emptyset$.

Proof: the separator is the opponent's mixture



Separation gives $q \neq 0$ with

$$q \cdot x \leq q \cdot y \quad (x \in X, y \in Y).$$

Let each coordinate of y grow: $q_j \geq 0$.

Normalize: $\sum_j q_j = 1$.

Let $y \downarrow (v, \dots, v)$:

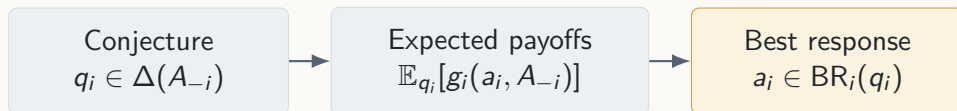
$$g(s_1, q) = q \cdot x \leq v.$$

$$\bar{v} \leq v = \underline{v} \quad \text{and hence equality.}$$

Lecture 2: Rationalizability

- 1 Minimax**
Guarantees; mixing; a separation argument
- 2 Rationalizability**
Best responses; conjectures; iterating rationality
- 3 Correlated equilibrium**
Signals; obedience; the revelation principle
- 4 Bayesian games**
Types; contingent choices; equilibrium

Best responses: rational choice needs a conjecture



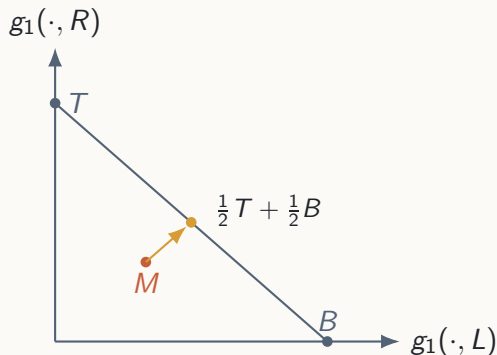
$$BR_i(q_i) = \arg \max_{a_i \in A_i} \sum_{a_{-i}} q_i(a_{-i}) g_i(a_i, a_{-i}).$$

$A_{-i} = \prod_{j \neq i} A_j$; opponents' actions may be correlated.

A mixture can dominate a pure action

	<i>L</i>	<i>R</i>
<i>T</i>	0	3
<i>M</i>	1	1
<i>B</i>	3	0

Only Player 1's payoffs.



$\frac{1}{2}T + \frac{1}{2}B$ yields $\frac{3}{2} > 1$ against both columns.

Strict dominance rules out a best response

$s_i \in \Delta(A_i)$ strictly dominates a_i if

$$g_i(s_i, a_{-i}) > g_i(a_i, a_{-i}) \quad \text{for every } a_{-i} \in A_{-i}.$$

Better against
every opponent profile



Better under
every conjecture

$$\mathbb{E}_{q_i}[g_i(s_i, A_{-i})] > \mathbb{E}_{q_i}[g_i(a_i, A_{-i})] \quad \text{for every } q_i.$$

Can every remaining action be justified?

Strictly dominated by a mixture $\iff$ never a best response.

Never a best response

$$\forall q_i \quad \exists b_i : \\ \mathbb{E}_{q_i}[g_i(b_i, A_{-i})] > \mathbb{E}_{q_i}[g_i(a_i, A_{-i})].$$

One dominating mixture

$$\exists s_i \quad \forall a_{-i} : \\ g_i(s_i, a_{-i}) > g_i(a_i, a_{-i}).$$

Minimax turns many local improvements into one uniform improvement.

Proof: build a game of payoff improvements

Fix the action a_i .

$$\gamma(b_i, a_{-i}) = g_i(b_i, a_{-i}) - g_i(a_i, a_{-i}).$$



best response,

$$\max_{b_i} \mathbb{E}_{q_i}[\gamma(b_i, A_{-i})] > 0 \quad \text{for every } q_i.$$

Continuity and compactness give a strictly positive minimum.

Proof: apply minimax once

$$\underbrace{\min_{q_i} \max_{b_i} \mathbb{E}_{q_i}[\gamma(b_i, A_{-i})]}_{\text{never a best response}} = \underbrace{\max_{s_i} \min_{a_{-i}} \gamma(s_i, a_{-i})}_{\text{one mixture works everywhere}} > 0.$$

Choose a maximizing s_i . Then

$$g_i(s_i, a_{-i}) - g_i(a_i, a_{-i}) > 0 \quad \text{for every } a_{-i}.$$

s_i strictly dominates a_i .

Conjectures: must opponents' actions be independent?

Player 1 chooses T, M, B ; Player 2 chooses L, R ; Player 3 chooses l, r .

Player 3 chooses l

	L	R
T	-8	0
M	0	0
B	-3	-3

Player 3 chooses r

	L	R
T	0	0
M	0	-8
B	-3	-3

Only Player 1's payoffs are displayed.

The same marginals can justify different choices

Independent conjecture

$$p = \Pr(L), q = \Pr(l).$$

$$\mathbb{E}[T] = -8pq,$$

$$\mathbb{E}[M] = -8(1-p)(1-q).$$

$$\min\{pq, (1-p)(1-q)\} \leq \frac{1}{4}.$$

$$\max\{\mathbb{E}[T], \mathbb{E}[M]\} \geq -2 > -3.$$

B is never a best response.

Correlated conjecture

	l	r
L	$\frac{1}{2}$	0
R	0	$\frac{1}{2}$

$$\mathbb{E}[T] = \mathbb{E}[M] = -4,$$

$$\mathbb{E}[B] = -3.$$

B is the unique best response.

At $p = q = \frac{1}{2}$, both conjectures have uniform marginals.

Iteration: what if I know that you are rational?

- 1 I am rational**
I choose a best response to some conjecture
- 2 I know that you are rational**
My conjecture excludes your unjustifiable actions
- 3 We keep reasoning about one another**
Restrict the conjectures again

Common knowledge of rationality motivates the whole iteration.

Two rounds of reasoning

	L	R
T	3, 2	2, 1
B	1, 0	0, 3

T strictly dominates B
Player 1 cannot choose B



Against T , Player 2 prefers L
Player 2 cannot choose R

The only surviving profile is (T, L) .

Delete, restrict the conjectures, repeat

$$R_i^0 = A_i, \quad R_{-i}^n = \prod_{j \neq i} R_j^n,$$

$$R_i^{n+1} = \{a_i \in R_i^n : \exists q_i \in \Delta(R_{-i}^n), a_i \in \text{BR}_i(q_i)\}.$$

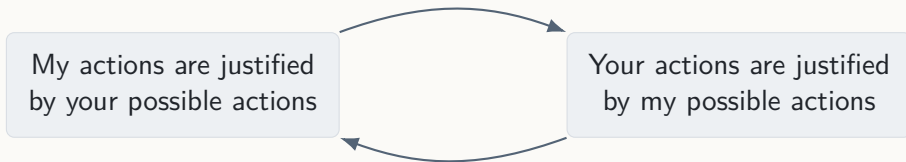


action sets: the decreasing sequence eventually stabilizes.

Rationalizability as mutual justification

A family $(B_i)_{i \in I}$ is jointly rationalizable if

$$\forall i, \forall b_i \in B_i, \exists q_i \in \Delta(B_{-i}) : b_i \in \text{BR}_i(q_i).$$



Take the largest such family: $\tilde{R}_i = \bigcup_{(B_j) \text{ jointly rationalizable}} B_i.$

Surviving deletion and mutual justification coincide

$$R_i = \tilde{R}_i \quad \text{for every player } i.$$

Survivors justify one another

At the stopping round,

$$R_i^{n+1} = R_i^n = R_i.$$

Every surviving action is a best response to a belief on R_{-i} .

A justified family survives every round

Start with $B_i \subseteq A_i$.

If $B_j \subseteq R_j^n$ for all j , each b_i has a justifying belief on R_{-i}^n .

Hence $B_i \subseteq R_i^{n+1}$.

Throughout: arbitrary joint conjectures about opponents.

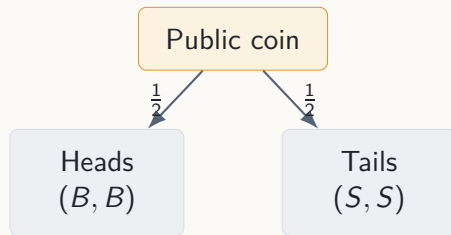
Lecture 2: Correlated equilibrium

- 1 Minimax**
Guarantees; mixing; a separation argument
- 2 Rationalizability**
Best responses; conjectures; iterating rationality
- 3 Correlated equilibrium**
Signals; obedience; the revelation principle
- 4 Bayesian games**
Types; contingent choices; equilibrium

Signals: a public coin can coordinate play

	B	S
B	2, 1	0, 0
S	0, 0	1, 2

Battle of the sexes.

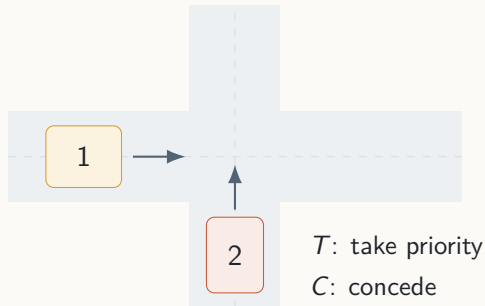


Both get $\frac{3}{2}$ versus $\frac{2}{3}$ in the mixed Nash equilibrium.

Can private signals do more?

	C	T
C	6, 6	2, 7
T	7, 2	0, 0

Chicken: concede or take priority.



Pure Nash equilibria: (T, C) and (C, T) .

Each driver sees only their own traffic signal

	R	G
R	$\frac{1}{3}$	$\frac{1}{3}$
G	$\frac{1}{3}$	0

Joint signal distribution.



C : concede



T : take priority

A recommendation changes the conditional belief about the other driver.

Why obey the signal?

Green: take priority

The other signal is red with probability 1.

$$\mathbb{E}[T \mid G] = 7, \quad \mathbb{E}[C \mid G] = 6.$$

Taking priority is optimal.

Red: concede

The other signal is red or green, each with probability $\frac{1}{2}$.

$$\mathbb{E}[C \mid R] = \frac{1}{2}(6 + 2) = 4,$$

$$\mathbb{E}[T \mid R] = \frac{1}{2}(7 + 0) = \frac{7}{2}.$$

Conceding is optimal.

The calculation is the same for both drivers.

Private recommendations improve both payoffs

$$\mu(C, C) = \mu(C, T) = \mu(T, C) = \frac{1}{3}, \quad \mu(T, T) = 0.$$

	Player 1	Player 2
Fair lottery over the pure Nash equilibria	$\frac{9}{2}$	$\frac{9}{2}$
Symmetric mixed Nash equilibrium	$\frac{14}{3}$	$\frac{14}{3}$
Private traffic signals	5	5

$$\mathbb{E}[g_i] = \frac{1}{3}(6 + 2 + 7) = 5.$$

Obedience: correlated equilibrium

Aumann (1974)

A mediator draws $a \sim \mu \in \Delta(A)$ and privately recommends a_i to player i .

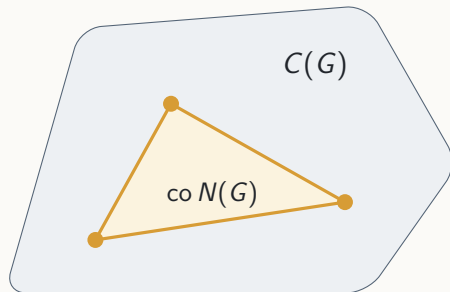
$$\sum_{a_{-i}} \mu(a_i, a_{-i}) [g_i(a_i, a_{-i}) - g_i(b_i, a_{-i})] \geq 0$$

for every i, a_i, b_i .

If $\Pr(a_i) > 0$:

$$\mathbb{E}[g_i(a_i, A_{-i}) \mid a_i] \geq \mathbb{E}[g_i(b_i, A_{-i}) \mid a_i].$$

A convex set of obedient distributions



Schematic

Obedience gives linear inequalities in μ .

$C(G)$ is a compact convex polytope.

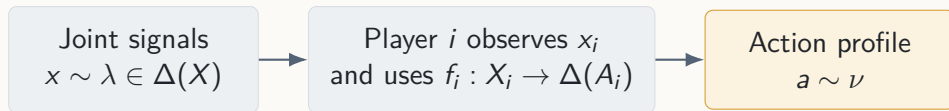
Every Nash distribution is obedient:

$$N(G) \subseteq C(G).$$

So are lotteries over Nash distributions:

$$\text{co } N(G) \subseteq C(G).$$

Revelation: start with arbitrary signals



$$\nu(a) = \sum_x \lambda(x) \prod_i f_i(a_i \mid x_i).$$

Payoffs depend on actions; the signals carry no payoff-relevant state.

The revelation principle for correlated equilibrium

Equilibrium with arbitrary private signals
 $\implies$ an obedient distribution of action recommendations.

Observe a signal x_i
then choose a_i



Receive recommendation a_i
then obey

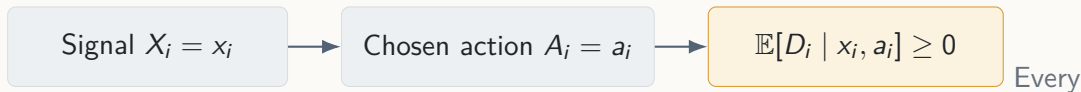
The action

distribution ν is preserved.

Proof: keep the signal while checking a deviation

Fix a player i , a recommended action a_i , and a deviation b_i .

$$D_i = g_i(a_i, A_{-i}) - g_i(b_i, A_{-i}).$$



action used at a signal is a best response.

Own randomization carries no extra information about opponents.

Proof: average over the signals behind a recommendation

$$\mathbb{E}[D_i \mid A_i = a_i] = \sum_{x_i} \Pr(x_i \mid A_i = a_i) \underbrace{\mathbb{E}[D_i \mid X_i = x_i, A_i = a_i]}_{\geq 0} \geq 0.$$

Best response
at each underlying signal

Average

Best response
given the recommendation alone

$\nu \in C(G)$ The same conditional-averaging idea as in Lecture 1.

Lecture 2: Bayesian games

- 1 Minimax**
Guarantees; mixing; a separation argument
- 2 Rationalizability**
Best responses; conjectures; iterating rationality
- 3 Correlated equilibrium**
Signals; obedience; the revelation principle
- 4 Bayesian games**
Types; contingent choices; equilibrium

Types: information can concern the payoffs

Information

States $\omega \in \Omega$.

Common prior $p \in \Delta(\Omega)$.

Player i observes the cell $P_i(\omega)$ of a partition $\mathcal{P}_i$.

Payoffs

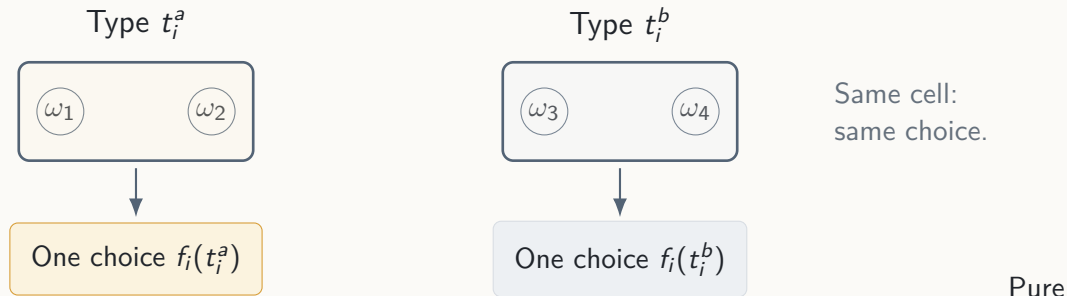
Payoff state $k = \kappa(\omega)$.

Actions $a_i \in A_i$.

Payoffs $g_i(a, \kappa(\omega))$.

A signal now changes beliefs about opponents *and* payoff-relevant states.

A strategy is a contingent plan



strategy: $f_i : T_i \rightarrow A_i$.

Behavioral strategy: $\sigma_i : T_i \rightarrow \Delta(A_i)$.

Do you really want to defect?

State L

	C	D
C	$0, -2$	$-10, -1$
D	$-1, -10$	$-5, -5$

State R

	C	D
C	$0, -2$	$-10, -7$
D	$-1, -10$	$-5, -11$

$$\Pr(L) = p, \quad \Pr(R) = 1 - p.$$

Player 1 chooses the row; Player 2 chooses the column.

First: nobody observes the state

	C	D
C	$0, -2$	$-10, 6p - 7$
D	$-1, -10$	$-5, 6p - 11$

Pure equilibrium (C, C)

$$p \leq \frac{5}{6}.$$

Pure equilibrium (D, D)

$$p \geq \frac{1}{6}.$$

Two pure equilibria coexist for $\frac{1}{6} \leq p \leq \frac{5}{6}$.

Now: only Player 2 observes the state

A plan specifies an action in L , then an action in R .

CC

CD

DC

DD

In state L : choose D

Against C : $-1 > -2$.

Against D : $-5 > -10$.

In state R : choose C

Against C : $-2 > -7$.

Against D : $-10 > -11$.

DC is optimal at each type, whatever Player 1 does.

Player 1 responds to the informed plan

Against DC :

$$\mathbb{E}[g_1(C, DC)] = -10p, \quad \mathbb{E}[g_1(D, DC)] = -4p - 1.$$

$$C \text{ is optimal} \iff p \leq \frac{1}{6}.$$



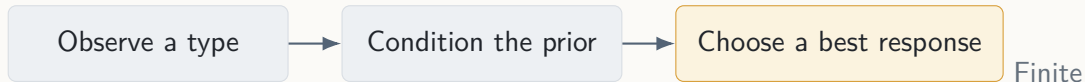
At $p = \frac{1}{6}$, Player 1 may use any mixture of C and D .

Equilibrium: best responses at every type

Write $\Omega = \prod_i T_i$; player i observes t_i .

A Bayesian Nash equilibrium satisfies, for every positive-probability type,

$$\sigma_i(t_i) \in \arg \max_{s_i \in \Delta(A_i)} \mathbb{E}_p[g_i(s_i, \sigma_{-i}(T_{-i}), \kappa(T)) \mid t_i] .$$



types and actions: existence follows from Nash's theorem.

Minimax and penalty kicks

von Neumann (1928)

Zur Theorie der Gesellschaftsspiele.

Chiappori, Levitt and Groseclose (2002)

*Testing Mixed-Strategy Equilibria When
Players Are Heterogeneous: The Case of
Penalty Kicks in Soccer.*

Rationality and correlation

Bernheim (1984); Pearce (1984)

Foundational papers on rationalizability.

Aumann (1974)

*Subjectivity and Correlation in Randomized
Strategies.*

Maschler, Solan and Zamir (2013), *Game Theory*.

Conclusion

Minimax

One mixture can guarantee a payoff

Rationalizability

Choices must survive reasoning about others

Correlated equilibrium

Private recommendations must be obedient

Bayesian games

Choices depend on information about the state

Lecture 3: how do we compare information, value it, and choose it?

Weak dominance: the order can matter

	L	R
T	1, 0	0, 2
M	0, 2	2, 0
B	1, 1	2, 1

Delete T first

Then L weakly dominates R .

Delete R , then M : (B, L) .

Delete M first

Then R weakly dominates L .

Delete L , then T : (B, R) .

The bound for independent conjectures

Cauchy–Schwarz:

$$\sqrt{pq} + \sqrt{(1-p)(1-q)} \leq \sqrt{p + (1-p)} \sqrt{q + (1-q)} = 1.$$

Hence

$$\min\{\sqrt{pq}, \sqrt{(1-p)(1-q)}\} \leq \frac{1}{2}.$$

$$\min\{pq, (1-p)(1-q)\} \leq \frac{1}{4}.$$

The full game of contingent plans

Only Player 2 is informed; first action in L , second action in R .

	CC	CD	DC	DD
C	$0, -2$	$10p - 10, 5p - 7$	$-10p, p - 2$	$-10, 6p - 7$
D	$-1, -10$	$4p - 5, p - 11$	$-4p - 1, 5p - 10$	$-5, 6p - 11$

DC is strictly dominant when $0 < p < 1$.

At $p = 0$ or $p = 1$, the plan at the zero-probability type is irrelevant.

Exercise: who knows the quality of the car?

Car values

	Seller	Buyer
Good car	3	4
Bad car	0	0

Each quality has probability $\frac{1}{2}$.

Trade at price π

Seller: sell or keep.

Buyer: buy or decline.

Trade requires both to agree.

Net gains: $(\pi - v_s, v_b - \pi)$.

No trade: payoff 0 for each.

Compare: both informed / neither informed / only seller informed.

For which prices can trade occur in equilibrium?

Why the maxmin never exceeds the minmax

For any s_1, s_2 ,

$$\min_{t_2} g(s_1, t_2) \leq g(s_1, s_2) \leq \max_{t_1} g(t_1, s_2).$$

Thus, for every s_2 ,

$$\max_{s_1} \min_{t_2} g(s_1, t_2) \leq \max_{t_1} g(t_1, s_2).$$

$$\max_{s_1} \min_{s_2} g(s_1, s_2) \leq \min_{s_2} \max_{s_1} g(s_1, s_2).$$